in multicellular organisms with relatively simple organisation. The offsprings show "uniparental inheritance" without any genetic variation. The different modes of asexual reproduction seen in animals are fission, budding, fragmentation and regeneration.

Fission is the division of the parent body into two or more identical daughter individuals. Five types of fission are seen in animals. They are **binary fission, multiple fission, plasmotomy, strobilation and sporulation.**

In **binary fission**, the parent organism divides into two halves and each half forms a daughter individual. The nucleus divides first amitotically or mitotically (karyokinesis), followed by the division of the cytoplasm (cytokinesis). The resultant offsprings are genetically identical to the parent. Depending on the plane of fission, binary fission is of the following types,

i) Simple irregular binary fission
ii) Transverse binary fission
iii) Longitudinal binary fission
iv) Oblique binary fission

Simple irregular binary fission is seen in *Amoeba* like irregular shaped organisms (**Fig. 1.1**), where the plane of division is hard to observe. The contractile vacuoles cease to function and disappear. The nucleoli disintegrate and the nucleus divides mitotically. The cell then constricts in the middle, so the cytoplasm divides and forms two daughter cells.

In **transverse binary fission**, the plane of the division runs along the transverse axis of the individual. e.g. *Paramecium* and *Planaria*. In *Paramecium* (**Fig. 1.2**) the macronucleus divides by amitosis and the micronucleus divides by mitosis.

In **longitudinal binary fission**, the nucleus and the cytoplasm divides in the longitudinal axis of the organism (**Fig 1.3**). In flagellates, the

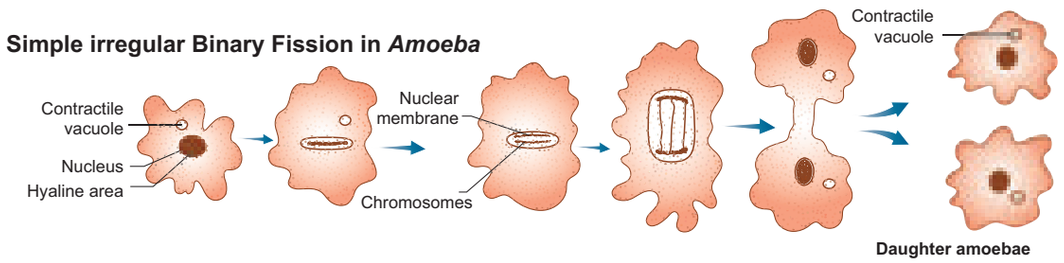


Fig. 1.1 Irregular simple binary fission in *Amoeba*

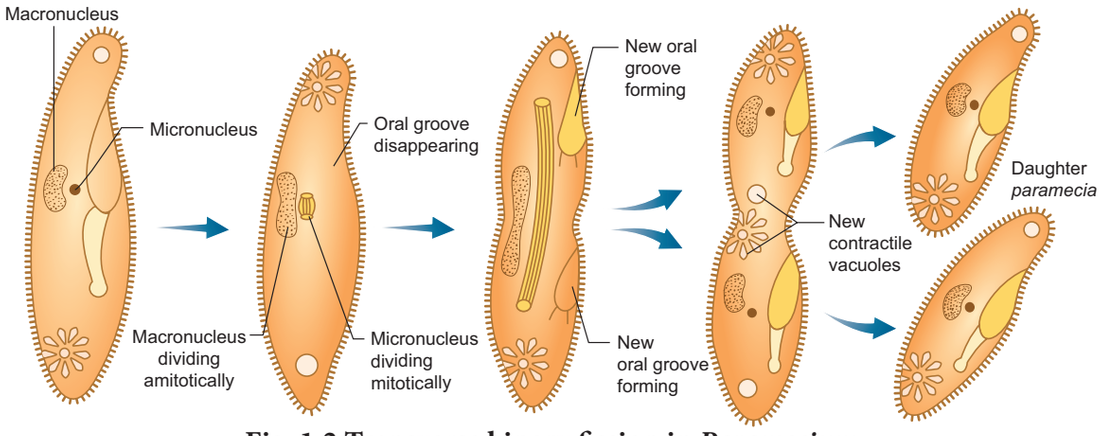


Fig. 1.2 Transverse binary fission in *Paramecium*

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